BIOCHEMISTRY CUMULATIVE EXAM

MARCH 4, 1995

A recent departmental seminar (fall, 1994) concerned the interaction of peptides with phospholipid membranes. Oriented samples were analyzed by the methods of X-ray diffraction and circular dichroism spectroscopy.

1. (15 points) Why would oriented samples be important for studies of peptide/membrane interactions?

2. a. (15 points) What physical principles underlie a circular dichroism experiment?

   b. (15 points) What is generally learned about proteins or peptides from a circular dichroism experiment?

   c. (15 points) Why would an oriented sample be used in a circular dichroism experiment? What additional information may be learned?

3. (20 points) What information could be learned from X-ray diffraction analysis of peptides with membranes? (Remember that these samples are <u>not</u> crystals!)

4. (20 points) If you remember the seminar, state two main conclusions from the research.
   Otherwise, propose results and conclusions that you might expect from research that uses the samples and methods described above.